

VocaNova 2025

Educational App

PROJECT TYPE Educational App	PLATFORM Native Android learning application
AUDIENCE Vocabulary learners using structured lessons	PRIMARY FOCUS Vocabulary practice and gamification

Overview

A comprehensive vocabulary learning Android application with interactive lessons, engaging mini-games, and gamification system. Features structured learning paths, achievement tracking, and immersive educational games with Material Design 3 interface.

Core Features

- 4-Week Structured Learning Program with Video Lessons
- Interactive Mini-Games (Flying Words, Synonym Matching)
- Gamification System with Coins & Power-ups
- Achievement Tracking & Progress Analytics
- Word of the Day with Audio Pronunciation
- Personal Vocabulary Library & Search
- Text-to-Speech Integration
- Offline Learning Capabilities

Experience and Design

- A four-week learning path combines video lessons, quizzes, and vocabulary review.
- Mini-games reinforce synonym recognition and rapid word recall.
- Coins, power-ups, achievements, and progress analytics encourage continued practice.
- Word of the Day, audio pronunciation, search, and saved words support daily review.

Architecture and Data

- Kotlin and Jetpack Compose provide the native Android interface.
- MVVM and Hilt separate UI state, use cases, repositories, and Firebase services.
- Firebase stores identity and synchronized learning progress.
- Media3 manages lesson video and audio playback with lifecycle-aware cleanup.

Security and Privacy

- Firebase Authentication protects account access.
- Firestore rules restrict learner data to its owning account.
- Validated writes reduce malformed lesson, reward, and progress records.
- Sensitive local values remain device-scoped and network traffic uses HTTPS.

Technology Stack

Kotlin	Application language
Jetpack Compose	Application framework and UI
Firebase	Authentication, database, or cloud data
Material Design 3	Interface styling and design system
Media3	Media storage and delivery
Hilt	State, data flow, and architecture

Performance and Reliability

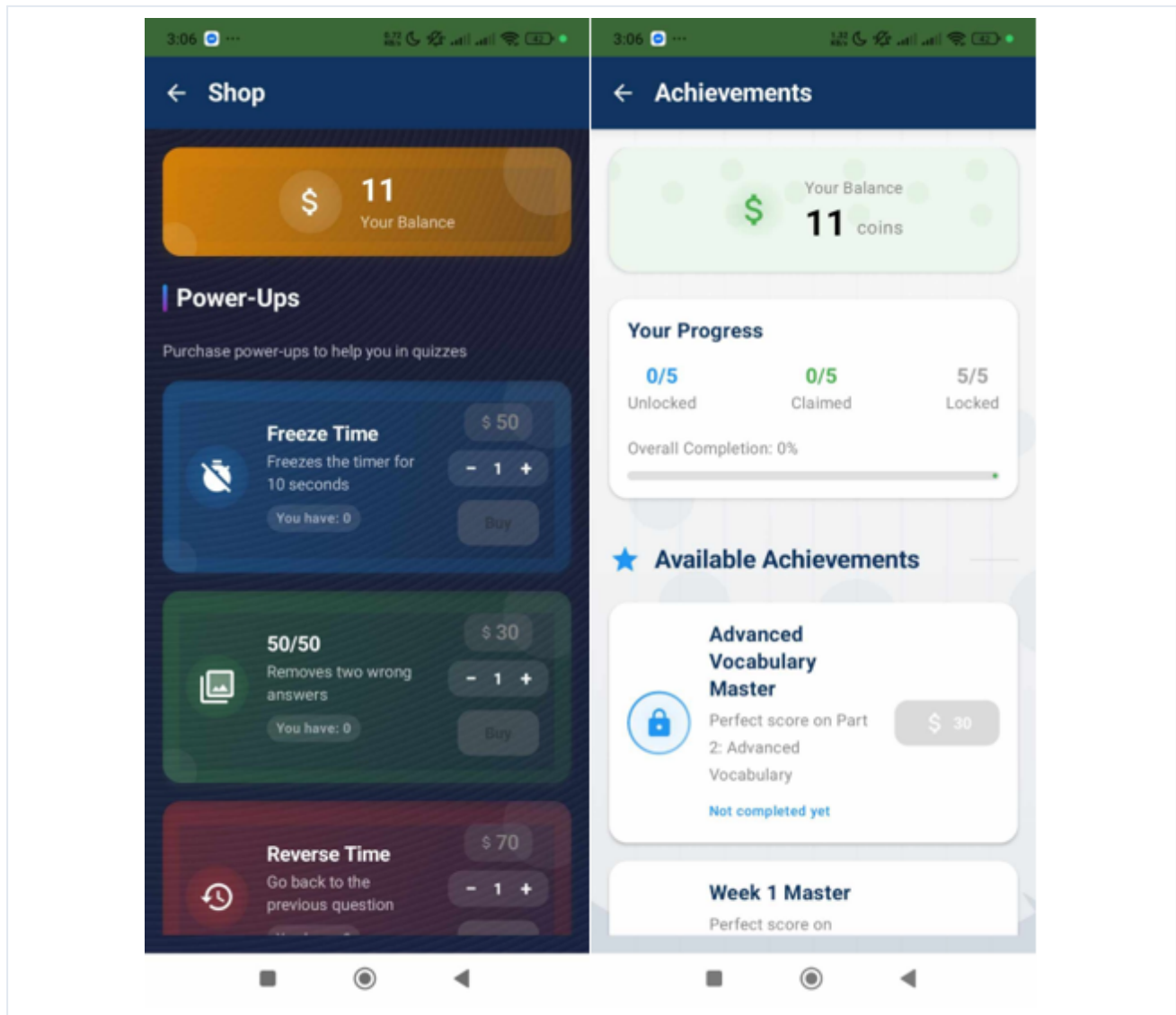
- Lazy Compose lists keep lesson, shop, and vocabulary collections responsive.
- State preservation maintains learner context through configuration changes.
- Local caching reduces repeated network access for vocabulary and progress data.
- Lifecycle-aware media management releases video and audio resources after use.

Platform Requirements

- Android 7.0 or later.
- At least 2 GB of RAM, with 4 GB recommended.
- Internet connection for authentication and cloud-synchronized progress.
- Audio output is recommended for pronunciation and lesson media.

PAGE OVERVIEW

Shop & Achievements

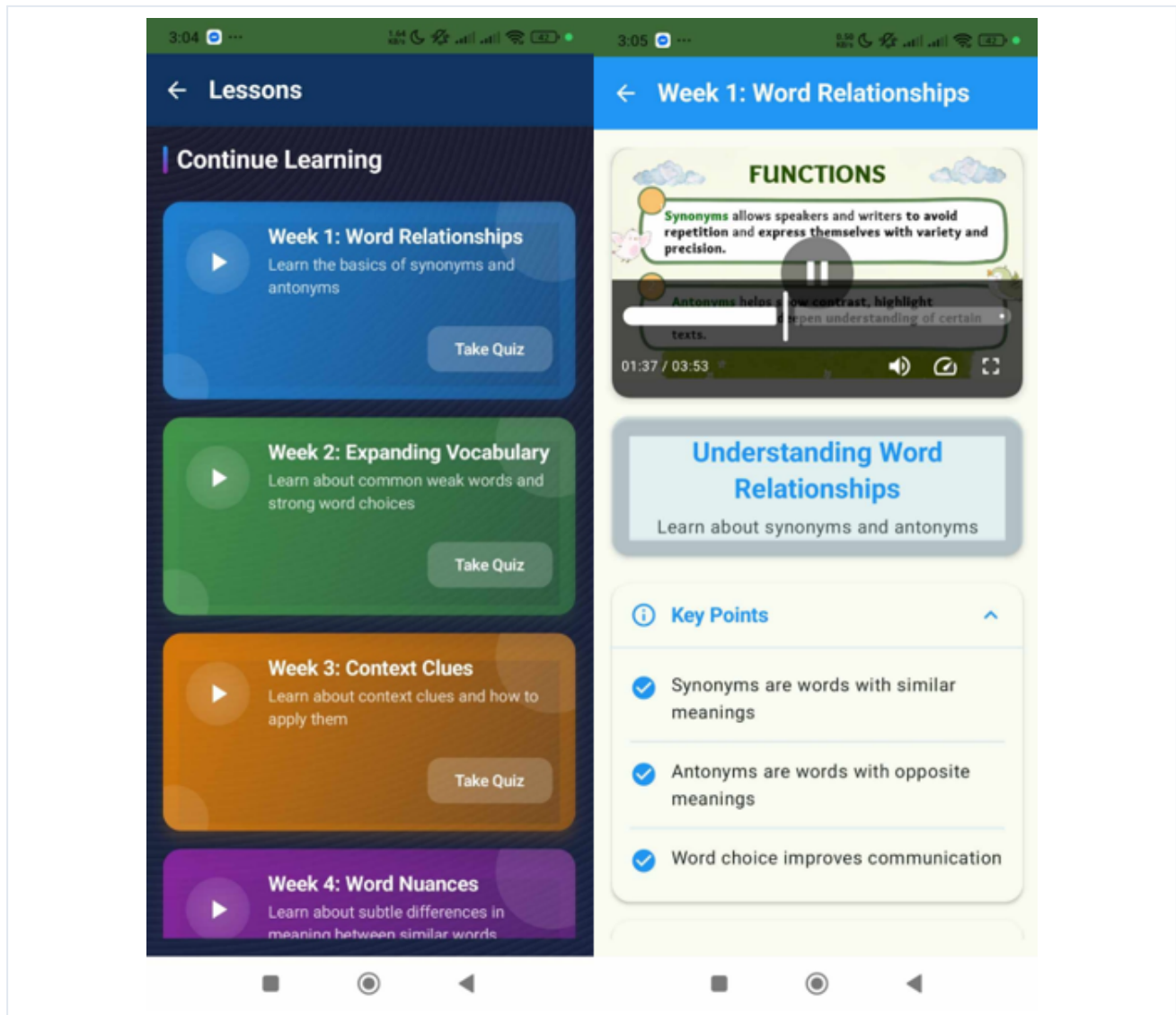


Shop & Achievements

Interface view from VocaNova 2025.

PAGE OVERVIEW

Lesson Screen

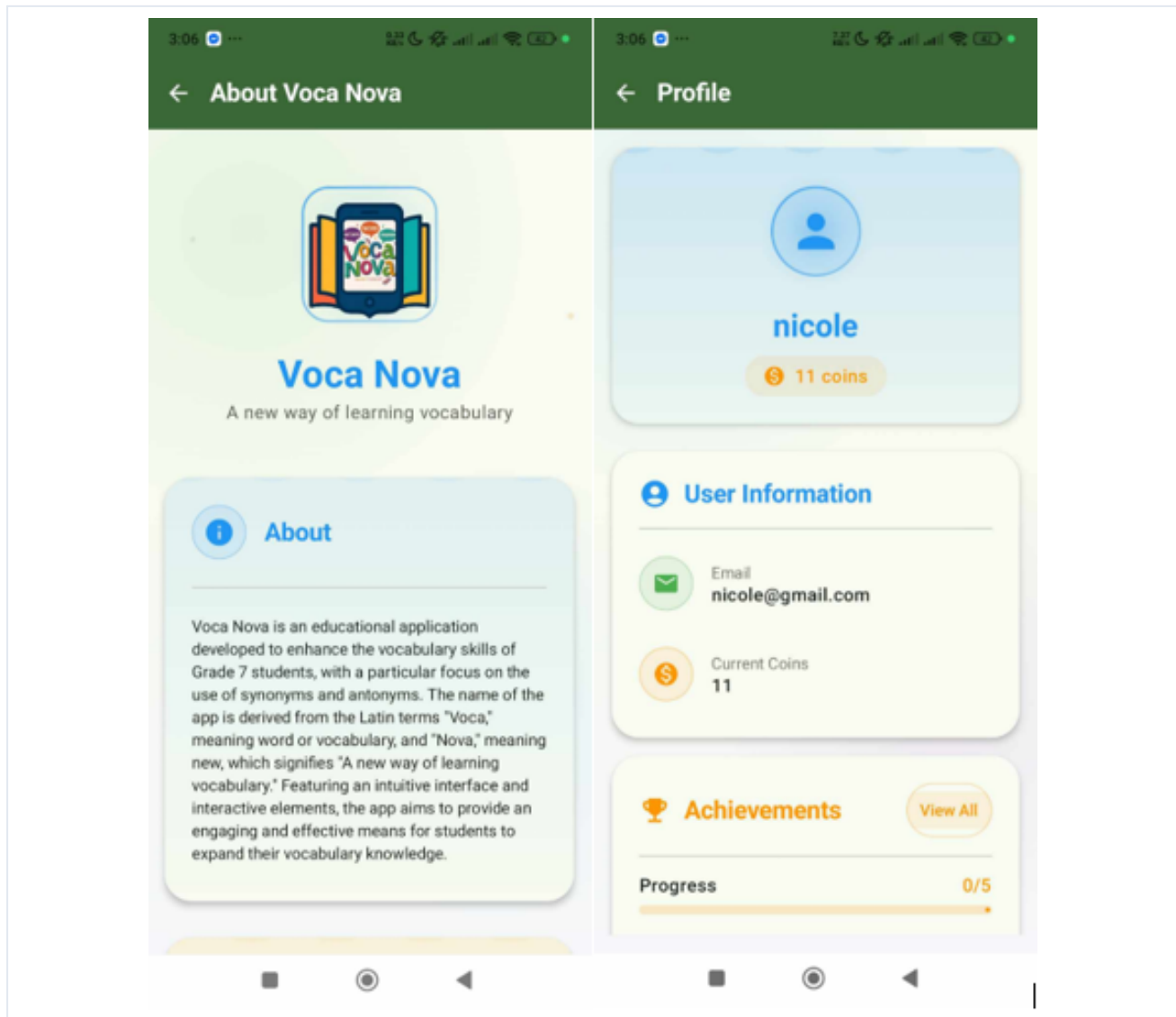


Lesson Screen

Interface view from VocaNova 2025.

PAGE OVERVIEW

About & User Info



About & User Info

Interface view from VocaNova 2025.

PAGE OVERVIEW

Quiz & Mini Game Screen

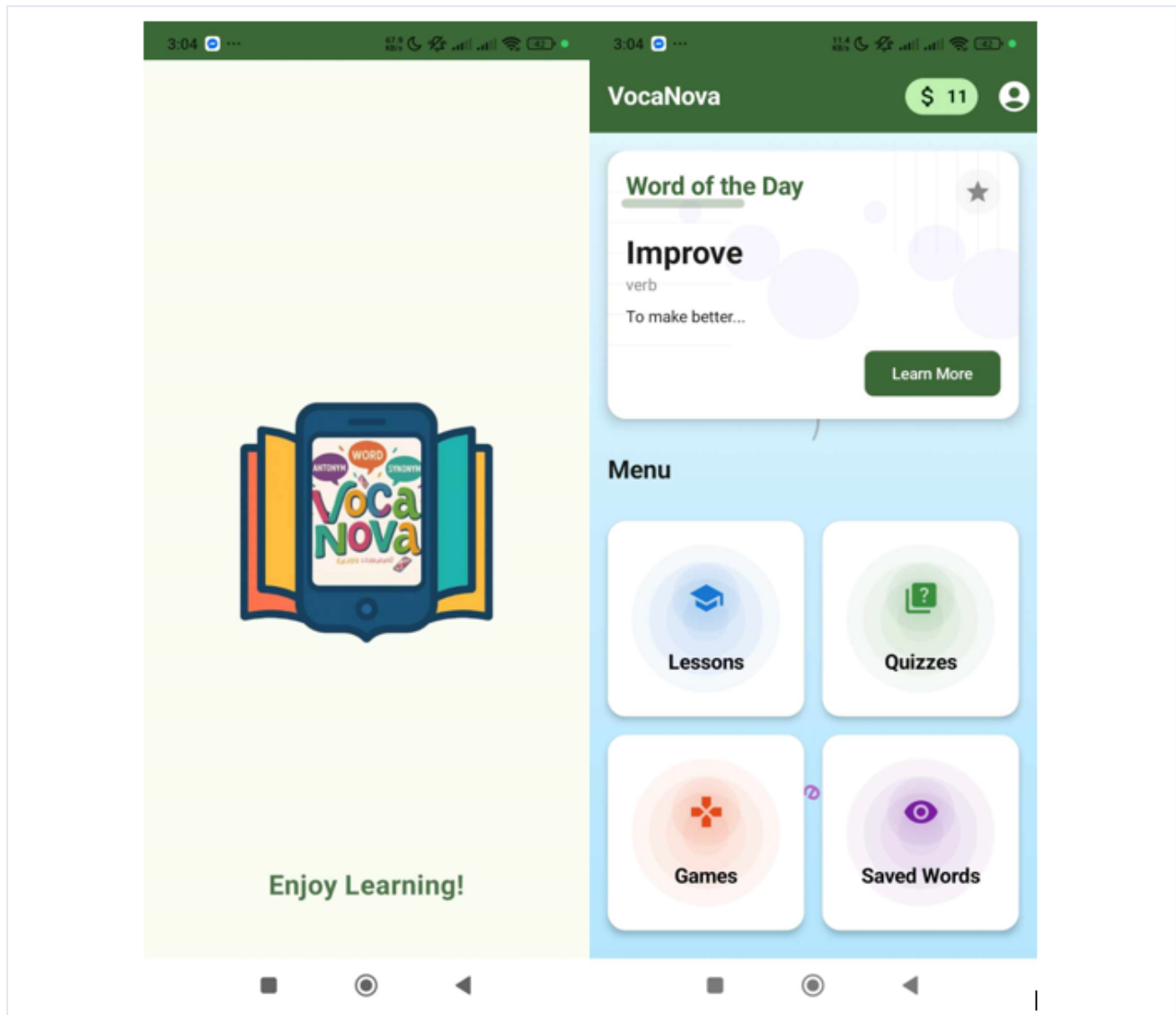


Quiz & Mini Game Screen

Interface view from VocaNova 2025.

PAGE OVERVIEW

Splash Screen & Home

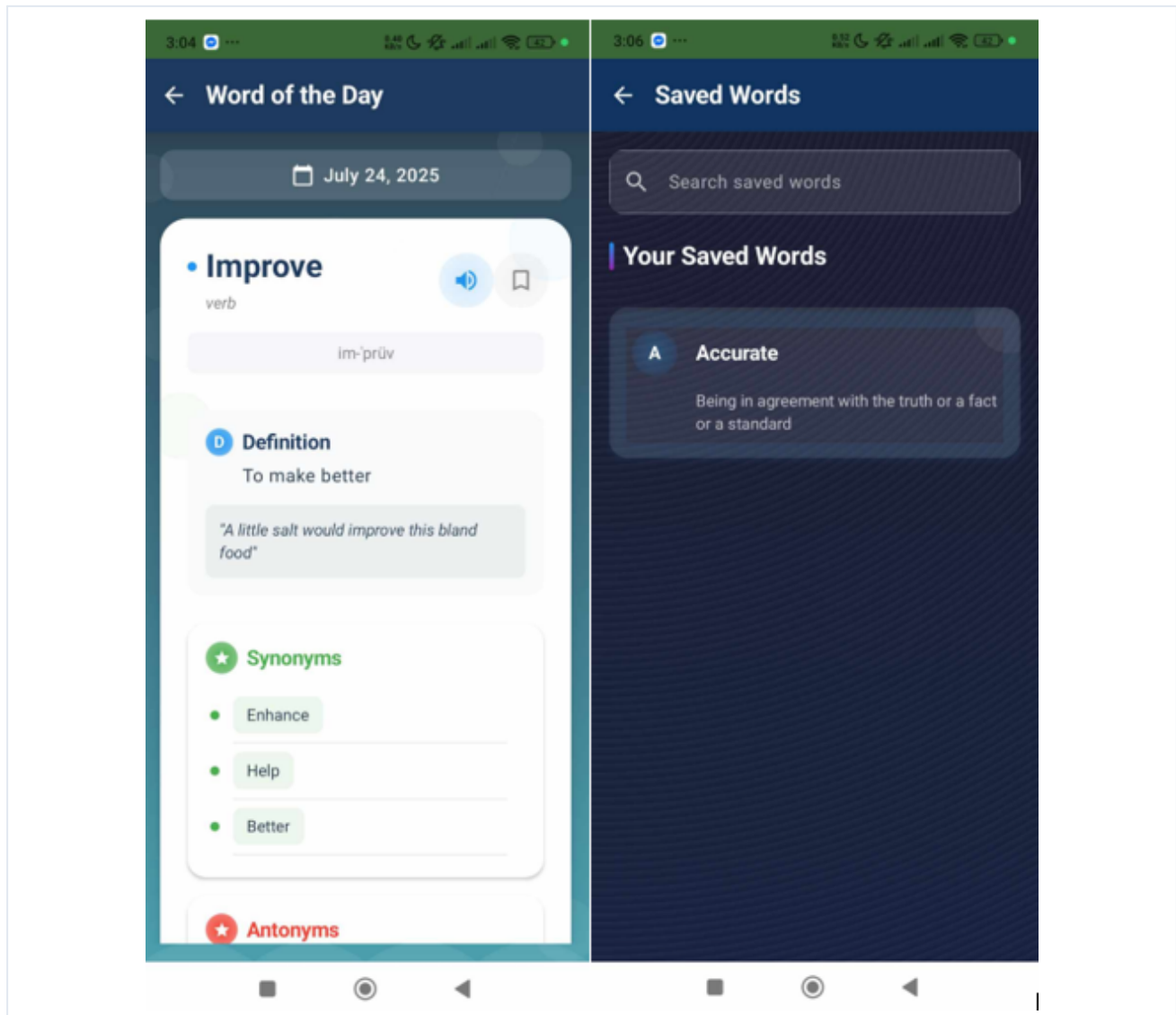


Splash Screen & Home

Interface view from VocaNova 2025.

PAGE OVERVIEW

Word of the Day & Saved Words



Word of the Day & Saved Words

Interface view from VocaNova 2025.

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Mini-Game Example



Mini-Game Example

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